Melissa, a medical student, noticed a small lump in her neck while putting on a necklace during her honeymoon. Following ultrasound imaging and biopsy of her thyroid, she was diagnosed with thyroid cancer and underwent thyroidectomy (surgical removal of the thyroid gland). During the procedure, Melissa's surgeon removed her entire thyroid gland while avoiding nearby blood vessels and nerves in her neck. In addition, the surgeon cut her parathyroid glands into smaller pieces and embedded them in the sternocleidomastoid muscles of her neck.

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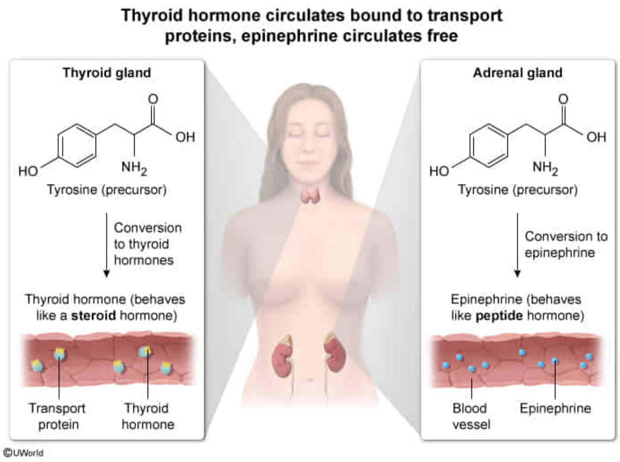
Although both thyroid hormone and epinephrine are synthesized from tyrosine, epinephrine behaves as a peptide hormone whereas thyroid hormone behaves like a steroid hormone. Given this, binding to a transport protein while in the circulation would likely occur for which hormone(s)?

- A. Thyroid hormone only [57%]
- B. Epinephrine only [34%]
- C. Both epinephrine and thyroid hormone [5%]
- D. Neither epinephrine nor thyroid hormone [1%]

Correct
56% answered correctly

Explanation:

Explanation:



Endocrine cells are found in various tissues throughout the body and secrete signaling molecules called hormones into the blood. Hormones can be broadly classified in several ways. For example, some hormones are steroid hormones, with a carbon skeleton derived from cholesterol, whereas others are composed of or derived from amino acids.

The question notes that both epinephrine and thyroid hormone are derived from the amino acid tyrosine and asks which hormone(s) likely circulates bound to a transport protein. Peptide hormones are water soluble and circulate largely in the free form (ie, unbound to proteins), whereas steroid hormones are lipophilic and are predominantly bound to proteins in the circulation. Because it is an amino acid-derived hormone that **behaves like a steroid hormone**, **thyroid hormone circulates bound** to a transport protein **(Choice D)**.

(Choices B and C) The question describes epinephrine as behaving like a peptide hormone; therefore, epinephrine circulates predominantly as a free (ie, *unbound*) molecule.

Educational objective:
Peptide hormones typically are hydrophilic, not bound to transport proteins in the blood, and dependent on cell membrane transporters to cross cellular membranes. Steroid hormones are lipophilic (being derived from cholesterol) and are typically bound to transport proteins in the blood but can diffuse across cellular membranes.

Time spent:0 sec
Subject:
Biology

QID:403963
System:
Endocrine and Nervous Systems

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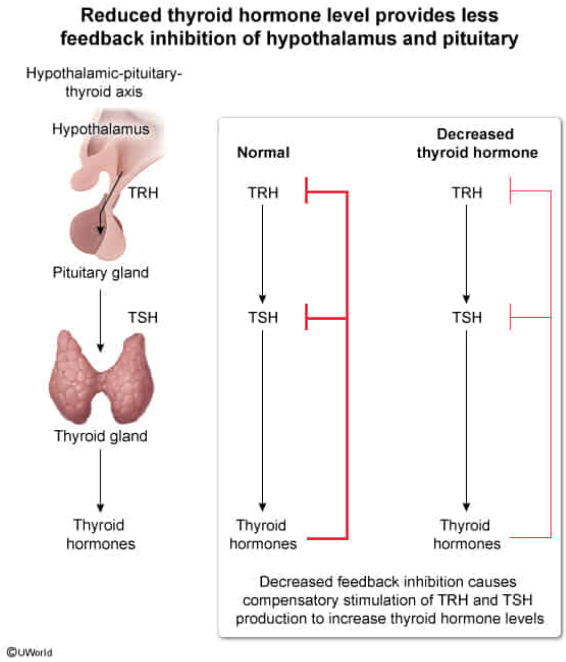
In the years before having her thyroid gland removed, a temporary decrease in Melissa's circulating thyroid hormone concentrations would have caused a compensatory response. Within the endocrine axis that includes thyroid hormone, which tissue(s) would respond to a temporary reduction in thyroid hormone concentration?

- A. Hypothalamus only [9%]
- B. Adrenal only [5%]
- C. Pituitary only [9%]
- D. Hypothalamus and pituitary [75%]

Correct
74% answered correctly

Explanation:



Hormones are signaling molecules secreted into the blood, typically in small amounts, that act upon receptors in target cells to elicit a response (eg, change in metabolism, homeostasis). Hormone release typically occurs in response to an initiating stimulus (eg, change in circulating metabolite concentration). One means of classifying hormones is according to whether they are regulated as part of an axis (ie, a self-regulating pathway of distinct tissues linked by their roles in the regulation of hormone release) or act alone.

The thyroid gland releases triiodothyronine (T_3) and tetraiodothyronine (T_4), two hormones that influence the function of most cells in the body. Most circulating thyroid hormone is T_4, but target tissues increase circulating T_3 levels by enzymatically converting T_4 to T_3, which is more potent. Activation of thyroid hormone receptors has a general stimulatory effect on many processes.

The passage describes a scenario in which Melissa—a medical student diagnosed with thyroid cancer—undergoes thyroid gland removal. The question asks which tissue is involved in the response to a decline in her circulating thyroid hormone concentration before the thyroidectomy. **Thyroid hormone production** is **regulated through** the linked activities of the hypothalamus, pituitary, and thyroid gland (ie, the **hypothalamic-pituitary-thyroid [HPT] axis**).

(Choices A and C) Both the hypothalamus and the pituitary are part of the HPT axis and will respond to a temporary decrease in thyroid hormone concentration.

(Choice B) In response to stress, the adrenal cortex releases cortisol due to endocrine stimulation via the hypothalamus and the pituitary gland. This cortisol release may influence the HPT axis, but cortisol is *not* itself part of the HPT axis.

Educational objective:
Thyroid hormone release is regulated by a self-regulating pathway that includes the linked activities of the hypothalamus, the pituitary, and the thyroid glands (ie, the hypothalamic-pituitary-thyroid axis).

Melissa, a medical student, noticed a small lump in her neck while putting on a necklace during her honeymoon. Following ultrasound imaging and biopsy of her thyroid, she was diagnosed with thyroid cancer and underwent thyroidectomy (surgical removal of the thyroid gland). During the procedure, Melissa's surgeon removed her entire thyroid gland while avoiding nearby blood vessels and nerves in her neck. In addition, the surgeon cut her parathyroid glands into smaller pieces and embedded them in the sternocleidomastoid muscles of her neck.

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Why were the pieces of Melissa's parathyroid gland reimplanted into her neck muscles by the surgeon during the thyroidectomy?

✔ ◉ A. To maintain normal osteoclast activity in bone [53%]

○ B. To avoid muscle and nerve problems caused by high blood calcium concentration [30%]

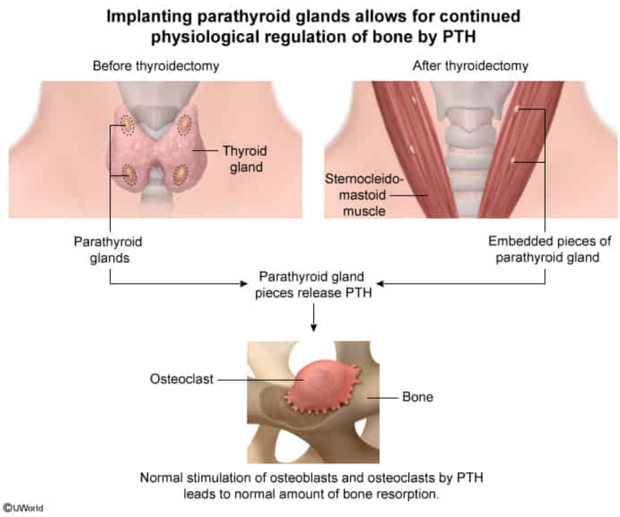
○ C. To prevent excessive blood calcium concentration due to inadequate urine calcium excretion [10%]

○ D. To promote phosphate reabsorption by the kidneys [5%]

Correct
52% answered correctly

Explanation:

Explanation:



Although 99% of the body's calcium is in the bone in the form of calcium phosphate crystals, calcium plays an integral role in processes throughout the body. Extracellular calcium concentrations are much higher than those in the cytosol of cells, and entry of calcium into the cytosol can depolarize (ie, increase the membrane potential) the cell. In addition, calcium entry into the cytosol from outside the cell or from intracellular compartments can act as a signal stimulating intracellular processes (eg, insulin release or muscle contraction).

The passage describes the reimplantation of the parathyroid glands in neck muscles during thyroidectomy (ie, surgical removal of the thyroid glands), and the question asks why this reimplantation was performed. The **parathyroid glands** play a **key role in** the **maintenance of normal blood calcium** concentration through the release of parathyroid hormone (PTH).

The role of PTH is to maintain an adequate calcium concentration in the blood, a task it accomplishes primarily by stimulating three processes: 1) intestinal absorption of calcium from the diet, 2) reabsorption of calcium filtered by the kidney (ie, reducing urinary calcium loss), and 3) promoting release of bone calcium into the blood.

Resorption of bone by osteoclasts is stimulated indirectly by PTH via osteoblasts. **PTH promotes production of osteoblasts**, as well as osteoblast release of ligands that promote differentiation of osteoclast precursor cells into osteoclasts. With **prolonged exposure to** elevated **PTH** concentrations in the blood, **osteoclast numbers increase**, leading to increased bone resorption.

(Choice B) In general, PTH contributes to maintenance of normal blood calcium levels by promoting entry of calcium into the blood. Therefore, reimplantation would not be performed for the purpose of lowering excessive blood calcium levels.

(Choice C) In the kidney, PTH stimulates calcium reabsorption by the body and phosphate excretion in the urine. Therefore, because PTH increases the calcium concentration of the blood, implantation of parathyroid glands would not prevent excessive blood calcium concentration.

(Choice D) PTH stimulates phosphate excretion (*not* reabsorption) by the kidneys.

Educational objective:
PTH stimulates osteoblasts to release ligands that stimulate differentiation of osteoclast precursor cells into osteoclasts. Persistent PTH elevation leads to increased bone resorption by osteoclasts.

Time spent: 0 sec	**QID**:403965
Subject: Biology	**System:** Endocrine and Nervous Systems

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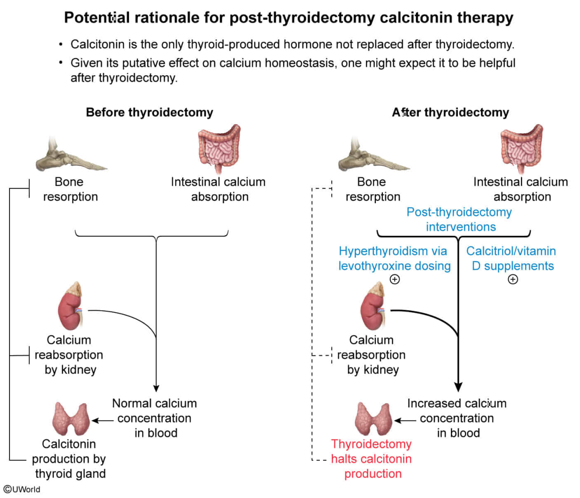
During a follow-up visit after her thyroidectomy, Melissa asked to be prescribed calcitonin along with her other medications and supplements. She likely made this request because:

I. Hyperthyroidism can lead to excessive stimulation of many processes, including bone resorption.
II. Calcitonin promotes iodine use for parathyroid hormone production.
III. Calcitonin is produced by the thyroid gland.
IV. Calcitriol is the most active form of vitamin D for stimulating intestinal calcium absorption.

- A. I and II only [10%]
- B. III and IV only [21%]
- C. I, II, and III only [20%]
- ✔ D. I, III, and IV only [47%]

Correct
47% answered correctly

Explanation:



Hormonal control of calcium levels in the body is thought to be carried out by primarily by calcitriol (produced by modifications of vitamin D in the liver and kidney), calcitonin (secreted by the C cells of the thyroid), and PTH.

The passage describes a patient who has her thyroid gland removed after being diagnosed with thyroid cancer and is prescribed replacement thyroid hormone. The patient requests that she start treatment with replacement calcitonin as well, and the question asks for the statement(s) that could likely have led to her making the request.

Several factors related to Melissa's thyroidectomy and her postsurgical treatment are relevant in consideration of whether she should receive calcitonin therapy:

- Thyroid hormones have a general stimulatory effect on many processes, and treatment with doses producing above-normal **thyroid hormone** concentrations **can overstimulate bone resorption** relative to bone deposition, **leading to reduced bone density**.
- **Calcitonin** is made in the **thyroid**, so it is reasonable to assume **it should be replaced** after thyroidectomy, like thyroid hormone.
- **Calcitriol** (vitamin D_3) and less active vitamin D supplements **stimulate intestinal calcium.absorption**. Given that calcitonin limits blood calcium concentration, one might reason that replacement calcitonin is necessary to prevent excessive calcium levels in the blood.

(Number II) Iodine is necessary for the synthesis of thyroid (*not* parathyroid) hormone.

Educational objective:
Calcitonin is secreted from the thyroid gland, and calcitonin is thought to limit blood calcium concentration by limiting calcium loss from bone. Thyroid hormones have a general stimulatory effect on most processes in the body, which can lead to imbalances if increased thyroid hormone concentrations stimulate one process to a greater degree than an opposing process. Calcitriol is the most active form of vitamin D and stimulates intestinal calcium absorption.

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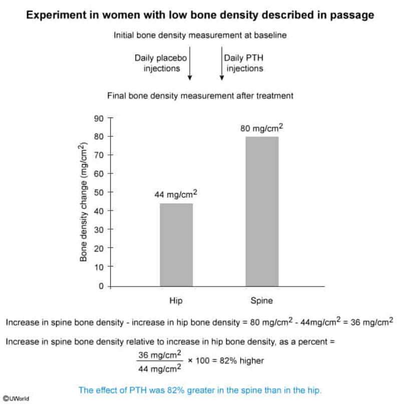
In the study that Melissa's physician discussed with her, how much more positive was the effect of PTH injection on spine bone density than the effect of PTH on hip bone density?

- A. 45% [29%]
- B. 55% [18%]
- ✔ C. 82% [38%]
- D. 180% [13%]

Correct
37% answered correctly

Explanation:



PTH's primary role as a mediator of calcium homeostasis is accomplished partly through its stimulation of osteoblast and osteoclast differentiation from their respective precursor cells. The effect on osteoblast differentiation is direct and is accompanied by the PTH-stimulated release of osteoblast factors that promote osteoclast differentiation. Osteoclasts, which do not possess PTH receptors, are thus indirectly stimulated to increase in numbers in response to PTH.

The question asks for the effect of daily PTH injections on spine bone density relative to the effect of daily PTH injections on hip bone density. The passage states that the PTH-induced increases in bone density were 80 mg/cm^2 for the spine and 44 mg/cm^2 for the hip. Therefore, the increase in spine bone density is 82%:

$$\frac{\text{Increase in spine bone density}}{\text{Increase in hip bone density}} = \frac{80\,\frac{\text{mg}}{\text{cm}^2}}{44\,\frac{\text{mg}}{\text{cm}^2}} = 1.82$$

Thus, in response to PTH injections, the **percent increase in spine** bone density is **82% more positive than** the percent increase **in the hip**.

(Choices A, B, and D) The values in these choices reflect incorrect calculations of the percent increase in spine bone density relative to that in the hip.

Educational objective:
PTH effects on bone density are determined by the relative stimulation of osteoblasts and osteoclasts.

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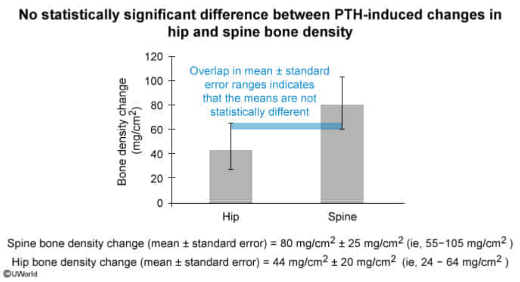
If the standard error for the increase in spine bone density is 25 mg/cm^2 and the standard error for the increase in hip bone density is 20 mg/cm^2, which of the following statements describes the PTH-induced changes in bone density in these two anatomic locations?

- A. The increases in spine and hip bone density do not differ significantly from one another. [57%]
- B. The increase in spine bone density is significantly greater than the increase in hip bone density. [20%]
- C. Bone density in the hip and spine were not significantly different at baseline. [7%]
- D. Bone density in the hip and spine were significantly different at baseline and after PTH treatment. [14%]

Correct
57% answered correctly

Explanation:



Experimental data are often summarized by expressing the results in terms of mean or median responses. Expressing data in these ways provides information about the overall responses to an intervention in different experimental groups. However, the variation in individual values in each group must be considered when evaluating whether group differences are real or likely due to chance.

In the passage, Melissa's physician describes a study in which PTH injections were associated with increases in bone density of the spine and hip of 80 mg/cm^2 and 44 mg/cm^2, respectively. The question adds the additional information that the standard error for the increase in spine bone density is 25 mg/cm^2 and the standard error for the increase in hip bone density is 20 mg/cm^2.

The question asks for the statement describing the PTH-induced changes in bone density in the spine and hip. The ranges defined by the means and standard errors for the spine and hip are:

$$\text{Spine bone density change}\left(\text{mean } \pm \text{ standard error}\right) = 80\ \frac{\text{mg}}{\text{cm}^2}\ \pm 25\ \frac{\text{mg}}{\text{cm}^2}\left(\text{ie, } 55 - 105\ \frac{\text{mg}}{\text{cm}^2}\right)$$

$$\text{Hip bone density change}\left(\text{mean } \pm \text{ standard error}\right) = 44\ \frac{\text{mg}}{\text{cm}^2}\ \pm 20\ \frac{\text{mg}}{\text{cm}^2}\left(\text{ie, } 24 - 64\ \frac{\text{mg}}{\text{cm}^2}\right)$$

In general, when the ranges defined by the means plus or minus *two* standard errors (which approximates the 95% confidence interval) for two groups overlap, they are not considered statistically different. For the current question, even adding or subtracting a single standard error to the means produces overlapping ranges. Because the **mean ± standard error ranges** for the spine and hip bone density changes **overlap**, they are **not** considered **statistically different**.

(Choice B) To be confident that two group means are significantly different (when statistical significance is not otherwise indicated), the ranges defined by the mean ± 2 standard errors must *not* overlap. Here, even the smaller mean ± standard error ranges for the spine and hip bone density changes overlap, so the means are not statistically different.

(Choices C and D) Without knowing the actual bone density values at baseline and after PTH treatment (rather than the changes from baseline presented in the passage), it is not possible to determine whether baseline differences in spine and hip bone density exist.

Educational objective:

When the mean ± standard error ranges for two groups overlap, they are considered similar (ie, not statistically different).

Time spent:0 sec
Subject:
Biology

QID:403968
System:
Endocrine and Nervous Systems